



Electrocardiogram 12 Lead Recording and Review Procedure

1. Scope

Site	Operational Area	Applicable to
Armadale Kalamunda Group	All services	Nursing and Medical

2. Purpose

The purpose of this procedure is to outline the requirements for recording and reviewing electrocardiogram (ECG) results by medical staff and nursing.

Key Points

- Good skin preparation and ECG electrode placement accuracy are essential to minimise interpretation errors.
- ECGs must have a minimum of three acceptable patient identifiers on the recording plus:
 - Reason for ECG (e.g. Pre-operative)
 - For chest pain - pain score
- An ECG completed for acute chest pain or as a result of clinical deterioration, must be reviewed by a senior medical officer (Registrar or Consultant) immediately within 10 minutes.
- Routine ECGs for reasons other than acute assessment must be reviewed by a medical officer within 24hours.
- Medical officers (MO) reviewing ECGs must sign and date the ECG before it is filed in the patient's medical record.
- Different areas within AKG use different equipment to record ECGs, staff need to familiarise themselves with the equipment in their area.

3. Procedure

- Explain the procedure and rationale for it (using AIDET) to the patient, family and / or carer and gain appropriate consent.
- Maintain patient privacy and dignity – only expose necessary body area.
- Maintain patient warmth.
- Remove watches and jewellery as necessary; ensure that these items remain with the patient.

3.1 Adult patients

- Expose lower legs and forearms for limb lead placement
- Position patient semi-fowler / supine on the bed with their arms relaxed at the sides and legs extended as clinical condition allows. (Note: Semi Fowlers position is defined as the head of a patient's bed raised approximately 25-30cm from base).
- If the bed / trolley is too narrow to support patient's limbs, consider placing patient's hands under their buttocks to prevent muscle tension.
- Expose the patient's chest.
- Ensure that skin is clean, dry and remove excessive hair.
- Whenever possible ECG electrodes should not be placed over healing surgical incisions, wounds, ulcers or abrasions.

3.2 Lead placement - adult limbs

- Choose sites over soft tissues.
- Avoid bony prominences, thick muscles or skin folds.
- Place ECG electrodes on patient's limbs with the lead connection pointing in a superior direction (towards torso).
- Arm leads to be equidistant along the arms and leg leads to be equidistant along the legs as able.
- If limbs not accessible (e.g. amputee or dressings), place electrodes at the most accessible distal location for that lead.
- **Right Leg (RL)** – on the inner aspect right leg (proximal to ankle).
- **Left Leg (LL)** – on the inner aspect left leg (proximal to ankle).
- **Right Arm (RA)** – on the inner aspect of the right forearm (proximal to wrist).
- **Left Arm (LA)** – on the inner aspect of the left forearm (proximal to wrist).

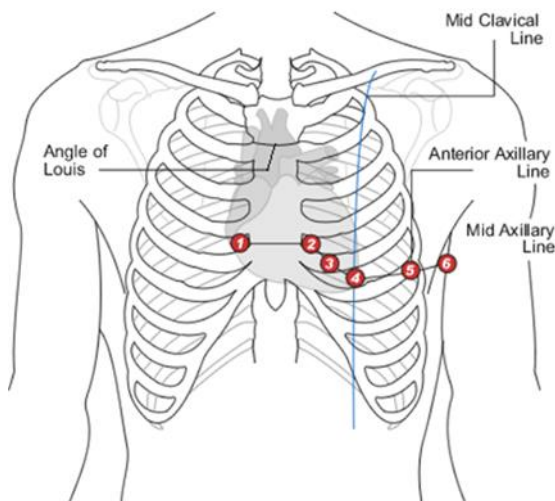
3.3 Lead placement - adult chest leads (precordial)

ECG cables to be placed on patient midline - ensure that the cables are not pulling on the electrodes.

Ensure that ECG cables are not twisted.

- **V1** – at fourth intercostal space, right sternal border
- **V2** – at fourth intercostal space, left sternal border
- **V3** – midway between V2 and V4
- **V4** – at the fifth intercostal space, left mid-clavicular line
- **V5** – at the same transverse level as V4 on anterior axillary line
- **V6** – at the same transverse level as V4 on the left mid axillary line

Figure 1: ECG chest lead placement



If placement of chest leads is modified (e.g. wound drain), label site with waterproof marker. If the patient has a large amount of breast tissue place the chest electrodes (V4-V6) underneath the breast tissue (displace the breast tissue as necessary).

Additional chest leads may be ordered as appropriate (e.g. V7, V8, V9) – seek guidance from senior staff.

3.4 Additional Lead placements

Adult right sided ECG

A right sided ECG should be performed if an acute inferior or posterior myocardial infarction is confirmed or suspected.

If a right sided ECG is requested, chest leads are placed in the same anatomical position on the right side of the chest (i.e. mirror image).

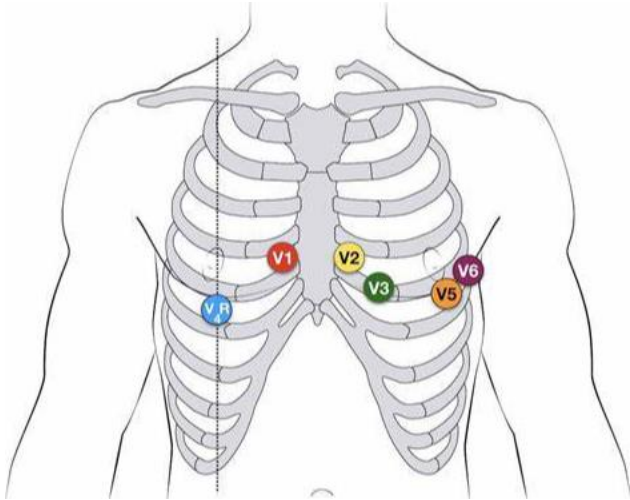
On the 12 lead ECG, **write 'R' next to leads on the right side** to identify a right-sided ECG.

Figure 2: Right-sided ECG placement



The most useful lead is **V_{4R}**, which is obtained by placing the V₄ electrode in the 5th right intercostal space in the mid-clavicular line. This can be obtained by moving only the V₄ electrode to the R side.

- ST elevation in V_{4R} has a sensitivity of 88%, specificity of 78% and diagnostic accuracy of 83% in the diagnosis of RV MI.



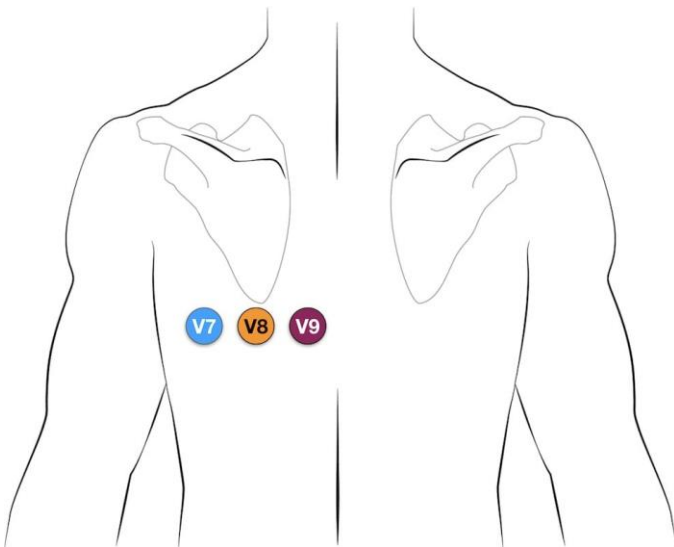
Posterior Leads

Be vigilant for evidence of posterior MI in any patient with an inferior or lateral STEMI.

To record a posterior lead ECG:

Leads V₇₋₉ are placed on the posterior chest wall in the following positions:

- **V₇** – Left posterior axillary line, in the same horizontal plane as V₆.
- **V₈** – Tip of the left scapula, in the same horizontal plane as V₆.
- **V₉** – Left paraspinal region, in the same horizontal plane as V₆.



3.5 Recording the 12 lead ECG

- Attach leads from 12 Lead ECG machine to corresponding electrode on patient.
- Instruct the patient relax and breathe normally whilst refraining from moving / talking.
- If respiratory movement interferes with an ECG recording, consider asking patient to briefly hold their breath/shallow breathe whilst the ECG is obtained.
- Ensure ECG cable is off the floor and clear of other electrical equipment and there is no tension on any of the leads.
- Observe the screen on the 12 lead ECG machine to ensure a clear signal is being obtained from each lead.
- Press ECG / Start to obtain ECG recording.
- ECG recording will commence once quality/steady baseline signal obtained
- Several factors can lead to electrical interference (artefact) or wandering baseline in the ECG recording – see troubleshooting below.

4. Paediatric patients

Ensure parental consent gained.

Child should be lying supine and preferably motionless.

If the child is anxious or distressed (< 5 years old), the parent may hold them ensuring they do not touch electrodes.

Consider using sucrose in infants < 3months

4.1 Lead placement - paediatric

Armada Kalamunda Group advocates standard adult lead placement. If paediatric electrodes not available ECG electrodes can be cut down longitudinally to an appropriate size to ensure electrodes are not overlapping.

5. After recording the ECG

5.1 Patient and equipment

- Gently remove all ECG electrodes and dispose in clinical waste.
- When serial ECGs are required within a two-hour time frame, the electrodes may be left in place on the patient.
 - Electrode contact, and position should be checked prior to performing subsequent ECGs.
- Replace patient clothing and items removed for ECG.
- Reposition patient as required to promote comfort.
- Wipe over ECG machine with 70% isopropyl alcohol wipes and refer to site specific infection control policy.
- Restock the ECG machine as required with paper and electrodes.
- Return the ECG machine to its correct location and ensure machine is plugged into AC power when not in use.

5.2 Documentation requirements

- Ensure each ECG is labelled **immediately** with 3 patient identifiers (Full name, DOB, UMRN) plus additional information.

- Reason for ECG (e.g. pre-operative, chest pain)
- Pain score
- Check the automatic date and time details, manually correct if not accurate.
- Ensure paper speed of 25mm/s and calibration spike of 10mm / mV.
- State any deviation from the standard lead position. e.g. V4R, V5R, V6R .
- Indicate any change in patient position from supine.
- Number ECG's when performed in a series for a single event / episode
- Document in the medical record
 - ECG for patient with chest pain / deterioration
 - Reason for ECG.
 - Time ECG performed.
 - Pain score if chest pain / discomfort is present and other related observations.
 - Time MO was informed.
 - Interventions and ongoing management plan.
 - ECG for routine assessment
 - Document ECG performed.

6. Troubleshooting

Problem	Possible Causes	Troubleshooting
Artefact (interference):	Patient experiencing seizures, chills or anxiety Improper electrode application Dry electrodes or fractured leads Electrical interference – electric beds, nurse call bells, electric foot-pumps, air mattresses and mobile phones	Keep the patient warm and encourage them to relax Ensure electrodes are in contact with the skin and connections secure, reapply as required Consider light abrasion of the skin surface with dry gauze wipe Ensure ECG cables are not twisted ECG wires may require replacement – refer to site based information Turn power supply to non-life support electrical equipment off
Weak or interrupted signal	Improper electrode attachment Lead failure	Ensure skin is clean and dry Ensure electrodes are in contact with the skin and connections secure, reapply as required ECG wires may require replacement – refer to site based information

Problem	Possible Causes	Troubleshooting
Wandering base line	Patient restless Chest wall movement during respiration Improper electrode application or positioning over bone	Sit patient up slightly Encourage patient to relax and breathe quietly Ensure that tension on the cable is not pulling the leads away from the patient's body during respiration Reposition electrodes

7. Legislative context

The following documents are required to give effect to this procedure:

- Health Department of Western Australia (HDWA) [Consent to Treatment Policy MP 0175/22](#)
- HDWA [Recognising and Responding to Acute Deterioration Policy MP 0171/22](#)
- East Metropolitan Health Service (EMHS) [Patient Identification and Procedure Matching EMHS: 21](#)

8. Supporting information

The following documents support the implementation of this procedure:

- EMHS [Consent Policy EMHS: 5](#)
- AKG [Clinical Documentation Policy. AK 0622/23](#)
- AKG [Clinical Handover Procedure. AK 0521/23](#)
- AKG [Recognition and Response to Acute Deterioration Policy. AK 0388/22](#)

9. Compliance, monitoring and evaluation

Compliance against this procedure will be evaluated through routine medical record audits overseen by the Deteriorating Patient and Resuscitation Committee.

10. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 2 – Action 2.3 Healthcare rights and informed consent
- Standard 6 – Action 6.5 Correct identification and procedure matching
- Standard 6 – Action 6.7 Clinical handover
- Standard 8 – Action 8.4 Recognising acute deterioration
- Standard 8 – Action 8.6 Escalating care
- Standard 8 – Action 8.10 Responding to deterioration

11. Policy owner (relating to these procedures)

Director Nursing and Midwifery

Enquiries relating to this procedure may be directed to:

Title: Nurse Coordinator – Medical services
Department: Hospital Wide
Email: jane.husain@health.wa.gov.au

12. Review

This procedure will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

Version	Effective from	Effective to	Amendment(s)
3	16/05/2024	16/05/2027	

13. Authorisation

This procedure has been authorised and issued by the [enter position title of the authoriser].

Authorised by	Helen Fullarton – Director Nursing and Midwifery
Authorisation date	16/05/2024
Published date	23/05/2024